Project: Worked examples
Title: Mechanism kinematics: four-bar, slider-crank, cam
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1. Scope and method

A single-degree-of-freedom planar mechanism driven at constant speed: a four-bar linkage or an offset slider-crank (both circuits, position, velocity and acceleration over a sweep of the input angle, Grashof class, transmission angle against a minimum, toggle and dead-centre positions, loop-closure residual, input torque by virtual work), or a cam with a translating follower. Rigid links, no clearance, no friction. Four-bar: $a\cdot e^{j\theta_2} + b\cdot e^{j\theta_3} = d + c\cdot e^{j\theta_4}$; $\ell = |A - O_4|$, $\cos\gamma = (c^2 + \ell^2 - b^2)/(2c\ell)$, $\theta_4 = \varphi \pm \gamma$ (φ the direction $O_4 \to A$; open: −, crossed: +); θ_3 from B − A Four-bar velocity and acceleration: differentiated loop, 2×2 system with $\det = b\cdot c\cdot\sin(\theta_3 - \theta_4)$; $\omega_3 = \omega_2\cdot a\cdot\sin(\theta_4 - \theta_2)/(b\cdot\sin(\theta_3 - \theta_4))$, $\omega_4 = \omega_2\cdot a\cdot\sin(\theta_3 - \theta_2)/(c\cdot\sin(\theta_3 - \theta_4))$; α from the second derivative at constant ω_2 Grashof: $s + l < p + q$ (crank-rocker, double-crank or double-rocker by the shortest link), = at the change point, > non-Grashof; transmission angle $\cos\mu = (b^2 + c^2 - \ell^2)/(2bc)$ folded to 0 … 90°; toggle where $\ell = b + c$ or $|b - c|$ Slider-crank: $x = r\cdot\cos\theta + \sigma\cdot\sqrt{l^2 - (r\cdot\sin\theta - e)^2}$ (σ = +1 open, −1 crossed); $dx/d\theta = -r\cdot\sin\theta - r\cdot\cos\theta\cdot(r\cdot\sin\theta - e)/(x - r\cdot\cos\theta)$; dead centres where crank and rod are collinear; $\mu = 90° - |$rod angle to the slide line$|$ Input torque by virtual work: $T_{in}\cdot\omega_{in} = \Sigma F\cdot v$; four-bar $T_{in} = (T_L + J\cdot\alpha_4)\cdot\omega_4/\omega_2$; slider $T_{in} = (m\cdot a + F_L)\cdot dx/d\theta$; RMS torque and mean power over the sweep by Simpson's rule Forces: four-bar tangential pin force $P = (T_L + J\cdot\alpha_4)/c$, coupler (and crank-pin) force $P/\sin\mu$; slider slide-direction force $F_x = m\cdot a + F_L$, rod force $F_x/\cos\theta_3$, guide reaction $F_x\cdot\tan\theta_3$; cam contact normal $F/\cos\varphi$ Assembly limits: four-bar toggles at $\cos\theta_2 = (d^2 + a^2 - \ell^2)/(2ad)$, $\ell = b + c$ or $|b - c|$; slider $\sin\theta = (e \pm l)/r$. The sweep is cut at them and, driven at constant speed into one, the unbounded peaks are not reported Loop closure: $|a\cdot e^{j\theta_2} + b\cdot e^{j\theta_3} - c\cdot e^{j\theta_4} - d|$ (slider: $|r\cdot e^{j\theta} + l\cdot e^{j\theta_3} - (x + j\cdot e)|$) asserted below 1e-9 mm at every step Cam, per segment with $x = (\theta - \theta_0)/\beta$: rise $s = s_0 + h\cdot f(x)$, return $s = s_0 - h\cdot f(x)$; cycloidal $f = x - \sin(2\pi x)/(2\pi)$; 3-4-5 $f = 10x^3 - 15x^4 + 6x^5$; 4-5-6-7 $f = 35x^4 - 84x^5 + 70x^6 - 20x^7$; modified trapezoid from its acceleration pieces ($C_v = 2$, $C_a = 8\pi/(\pi + 2)$, $C_j = 32\pi^2/(\pi + 2)$) Cam SVAJ at constant ω: $v = \omega\cdot s'$, $a = \omega^2\cdot s''$, $j = \omega^3\cdot s'''$ ($' = d/d\theta$); peak coefficients $C_v = \max f'$, $C_a = \max|f''|$, $C_j = \max|f'''|$, so $v_{pk} = C_v\cdot h\cdot\omega/\beta$ Pressure angle (roller, offset e, $R_p = R_b + R_r$): $\varphi = \arctan[(s' - e)/(\sqrt{R_p^2 - e^2} + s)]$, which for $e = 0$ is $\arctan[s'/(R_p + s)]$; the base circle that holds $\varphi \le \varphi_{max}$ has $R_{p,req} = \sqrt{s_{o,req}^2 + e^2}$ with $s_{o,req} = \max(|s' - e|/\tan\varphi_{max} - s)$ Radius of curvature of the pitch curve $P(\theta) = \mathrm{Rot}(-\theta)\cdot(e, \sqrt{R_p^2 - e^2} + s)$: $\rho = |P'|^3/(P' \times P'')$, convex where positive; undercut where $\rho_{min} \le R_r$ (FAIL), none where the curve is nowhere convex; flat-faced follower: signed profile radius $\rho = R_b + s + s''$, the minimum of which must stay above zero (FAIL at or below) and the face must reach $|s'|$; the cam turns counter-clockwise with the follower axis at +x offset e Cam follower force $F = m\cdot a + k\cdot s + F_{pre} + F_{ext}$ (the follower leaves the cam where $F < 0$); input torque by virtual work $T = F\cdot s'$ ($T\cdot\omega = F\cdot v$)

Ref: Norton, Design of Machinery Ch. 4, 6, 7 — Position, velocity and acceleration analysis; transmission angle; Grashof condition
Ref: Uicker, Pennock and Shigley, Theory of Machines and Mechanisms Ch. 4–6 — Loop-closure position analysis; velocity and acceleration of planar linkages

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mode	Kind of mechanism	Cam		spec: Indexing station IDX-12 §2 rev A	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
e_off	Offset e: slide line above the crank centre (slider-crank; blank: 0); cam follower axis offset, with the cam turning counter-clockwise and the follower axis at +x: positive e reduces the pressure angle where ds/dθ exceeds e (the rise), negative raises it	10	mm	drawing: IDX-12-CAM rev A	-1000 … 1000 mm
steps	Intervals N of the table (N + 1 rows, a whole number, at most 3600; the peaks do not depend on it)	72		assumption: 5° table step.	4 … 720
n_in	Input speed (constant)	120	rpm	spec: Indexing station IDX-12 §2: 120 index cycles/min rev A	0.1 … 30000 rpm
follower	Follower type: roller (translating, pressure angle and undercut checked) or flat-faced (in-line)	Roller		drawing: IDX-12-CAM rev A	any
Rb	Base-circle radius of the cam profile R_b	40	mm	drawing: IDX-12-CAM rev A	1 … 1000 mm
R_r	Roller radius R_r (roller follower)	8	mm	drawing: IDX-12-CAM rev A	1 … 200 mm
phi_max	Maximum pressure angle allowed (roller follower)	30	deg	assumption: Usual limit for a translating roller follower; set it from your design rule.	5 … 45 deg
segs	Cam segments: rise, dwell or return, with motion law, angle and lift; the angles add to 360° and the rises equal the returns	4 row(s)		spec: Indexing station IDX-12 §2.3 motion table rev A	any
segs[1]		kind = Rise; law = Cycloidal; beta = 120 deg; h = 20 mm			
segs[2]		kind = Dwell; law = –; beta = 60 deg; h = 0 mm			

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
segs[3]		kind = Return; law = 3-4-5 polynomial; beta = 120 deg; h = 20 mm			
segs[4]		kind = Dwell; law = —; beta = 60 deg; h = 0 mm			
m_f	Follower mass (all moving parts that follow the cam)	0.2	kg	assumption: Example: follower arm, roller and tooling; take it from the CAD mass properties.	0 … 1000 kg
k_sp	Return spring rate (link M7 rate k)	5	N/mm	assumption: Example return-spring rate; link the rate k of the M7 spring sheet.	0 … 1000 N/mm
F_pre	Spring preload at the base circle (link an M7 force)	20	N	assumption: Example preload at the base circle; link a force of the M7 spring sheet.	0 … 100000 N
F_ext	External load on the follower, positive against the rise	0	N	assumption: No external load on the follower in this example.	-100000 … 100000 N

3. Assumptions

A1. Rigid links, a planar mechanism with one degree of freedom, no joint clearance and no friction; the input turns at a constant speed. (Addendum B §B6 M9 limits.)
A2. The position is the closed-form solution of the vector loop; the open and crossed circuits are the two branches ($\theta_4 = \varphi - \gamma$ and $\varphi + \gamma$ for the four-bar, $\sigma = +1$ and -1 for the slider) and the sweep stays on the chosen branch; for a change-point linkage (parallelogram, deltoid) the circuit is named at a reference angle that does not depend on the sweep (the middle of the first interval between change points in 0 … 2π, π/2 when there are none), so the label at a crank angle means the same motion on every turn; only the true change points (all four links collinear, $\theta_2 = 0$ or π) flip the branch, and it is followed by continuity through them inside the sweep. A linkage with one change point per turn (a kite) repeats every 720° and a sweep of more than one turn warns. Angles at which the linkage cannot be assembled, or at a toggle where the velocity system is singular, are left out of the table and counted. (Norton, Design of Machinery, position analysis of the four-bar and the offset slider-crank.)
A3. Input torque is by virtual work from the output load and the inertia of the output element only (four-bar: the output link's moment of inertia about its pivot; slider: the slider mass); the crank and the coupler or connecting rod are massless. (No link-inertia or gravity terms: add the link masses to the output element

for a first allowance.)
A4. The peaks (speeds, accelerations, torque, transmission angle, swing) are those of the continuous motion over the swept range, found by a dense scan refined by golden section; the printed table (steps N, up to 3600) only samples it. (Peaks do not depend on the table's step count.)
A5. Cam: a translating follower on a cam turning at constant speed; the follower mass, a linear return spring (rate and preload at the base circle, for example linked from M7) and a constant external load make the force the cam must apply, F = m·a + k·s + F_pre + F_ext; contact is lost where it falls below zero. The segment table must add to 360° and close; pressure angle and undercut are checked for the roller follower (R_p = R_b + R_r), the positive profile radius for the flat-faced one. (Norton, Design of Machinery, cam design: the four classical laws, pressure angle and radius of curvature.)
A6. Follower force (four-bar: the tangential force at the output pin, $(T_L + J\cdot\alpha_4)/c$; slider: the slide-direction force m·a + F_L; cam: the axial force) is the force the mechanism must apply to the output element. The force in the coupler (F_pin_pk) is the tangential force over sin μ; in the connecting rod it is the slide-direction force over cos θ_3, and the guide reaction is that force times tan θ_3; the roller contact normal force is F/cos φ. Crank, coupler and rod are massless two-force members. (Rigid body, frictionless.)
A7. When the sweep meets an assembly limit (a toggle of a four-bar, the rod perpendicular to the slide line of a slider-crank) the sweep is cut there and, because the speed, acceleration, torque and forces at constant input speed are unbounded at the limit, they are not reported; the limit angles are listed. A change-point four-bar (parallelogram, deltoid) is continued through its singular position on the same branch by continuity (γ changes sign there), but its speeds and torques next to the singular position are not reported. (Closed-form limit angles: $\ell = b + c$ or $|b - c|$ (four-bar), $|r\cdot\sin\theta - e| = l$ (slider).)
A8. Into F1 and E1: output_motion is a kinematic proxy at constant crank speed, not the cylinder speed of an actuated linkage. For E1 use the peak input_torque for the peak check and T_rms_in with n_in for the thermal sizing; the mean power is P_mean. (Chains 4 and 5, Addendum B §B7.)
A9. steps (Intervals N of the table (N + 1 rows, a whole number, at most 3600; the peaks do not depend on it)) is taken as 72. (5° table step.)
A10. phi_max (Maximum pressure angle allowed (roller follower)) is taken as 30 deg. (Usual limit for a translating roller follower; set it from your design rule.)
A11. m_f (Follower mass (all moving parts that follow the cam)) is taken as 0.2 kg. (Example: follower arm, roller and tooling; take it from the CAD mass properties.)
A12. k_sp (Return spring rate (link M7 rate k)) is taken as 5 N/mm. (Example return-spring rate; link the rate k of the M7 spring sheet.)
A13. F_pre (Spring preload at the base circle (link an M7 force)) is taken as 20 N. (Example preload at the base circle; link a force of the M7 spring sheet.)
A14. F_ext (External load on the follower, positive against the rise) is taken as 0 N. (No external load on the follower in this example.)

4. Working

[15] theta_sw = □ — Cam follower kinematics
= 73 rows, □ from 0° to 360°
= [0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, … (73 items)] deg
Table: cam angle

[16] s_sw = s = s_0 + h·f(□/□) — Cam follower kinematics
= 73 rows, □ from 0° to 360°
= [0, 0.0094867, 0.075117, 0.24921, 0.57669, 1.092, 1.8169, 2.7587, 3.91, 5.2492, 6.7418, 8.3 items)] mm
Table: follower displacement

[17] v_sw = v = □·ds/d□ — Cam follower kinematics
= 73 rows, □ from 0° to 360°
= [0, 0.0040889, 0.016077, 0.035147, 0.06, 0.088942, 0.12, 0.15106, 0.18, 0.20485, 0.22392, 0 (73 items)] m/s
Table: follower velocity

[18] a_sw = a = □²·d²s/d□² — Cam follower kinematics
= 73 rows, □ from 0° to 360°
= [0, 1.1709, 2.2619, 3.1989, 3.9178, 4.3697, 4.5239, 4.3697, 3.9178, 3.1989, 2.2619, 1.1709,
m/s²
Table: follower acceleration

[19] j_sw = j = □³·d³s/d□³ — Cam follower kinematics
= 73 rows, □ from 0° to 360°
= [170.55, 164.74, 147.7, 120.59, 85.273, 44.141, 1.0443e-14, -44.141, -85.273, -120.59, -147
(73 items)] m/s³
Table: follower jerk

[20] phi_sw = □ = arctan[(s' − e)/(√(R_p² − e²) + s)] — Cam follower kinematics
= 73 rows, □ from 0° to 360°
= [-12.025, -11.642, -10.507, -8.6775, -6.2746, -3.4811, -0.52955, 2.3281, 4.8597, 6.8841, 8.
… (73 items)] deg
Table: pressure angle

[21] kappa_sw = □ = 1/□ of the pitch curve (positive convex, negative concave) — Cam follower kinematics
= 73 rows, □ from 0° to 360°
= [0.020833, 0.01767, 0.01465, 0.011975, 0.0098989, 0.0086679, 0.008423, 0.0091267, 0.010569,
0.012452, 0.014489, 0.016465, … (73 items)] 1/mm
Table: curvature of the pitch curve (the radius of curvature is 1/κ where κ is positive)

[22] F_sw = F = m·a + k·s + F_pre + F_ext — Cam follower kinematics
= 73 rows, □ from 0° to 360°
= [20, 20.282, 20.828, 21.886, 23.667, 26.334, 29.989, 34.667, 40.334, 46.886, 54.161, 61.948
items)] N
Table: follower force on the cam

[23] T_in_sw = T = F·ds/d□ — Virtual work
= 73 rows, □ from 0° to 360°
= [0, 0.0065993, 0.026647, 0.061213, 0.113, 0.18639, 0.28638, 0.41673, 0.57774, 0.76432, 0.96
… (73 items)] N·m
Table: input torque by virtual work

[24] seg_start = □ □ of the segments before — Cam follower kinematics
= 4 segments
= [0, 120, 180, 300] deg
Segments: start angle

[25] seg_beta = □ — Cam follower kinematics
= 4 segments
= [120, 60, 120, 60] deg
Segments: angle

[26] seg_s_end = s at the end of the segment — Cam follower kinematics
= 4 segments
= [20, 20, 0, 0] mm
Segments: displacement at the end

[27] seg_v_pk = max |v| in the segment — Cam follower kinematics
= 4 segments
= [0.24, 0, 0.225, 0] m/s
Segments: peak velocity

[28] seg_a_pk = max |a| in the segment — Cam follower kinematics
= 4 segments
= [4.5239, 0, 4.1569, 0] m/s²
Segments: peak acceleration

[29] seg_j_pk = max |j| in the segment — Cam follower kinematics
= 4 segments
= [170.55, 0, 259.2, 0] m/s³
Segments: peak jerk

[30] lift = max s — Cam follower kinematics
= over the revolution
= 20 mm
Greatest follower displacement

[31] v_pk = max |v| — Cam follower kinematics
= scan of each segment, refined
= 0.24 m/s
Peak follower velocity

[32] a_pk = max |a| — Cam follower kinematics
= scan of each segment, refined
= 4.5239 m/s²
Peak follower acceleration

[33] j_pk = max |j| — Cam follower kinematics
= scan of each segment, refined
= 259.2 m/s³
Peak follower jerk

[34] F_max = max F — Cam follower kinematics
= scan of each segment, refined
= 120 N
Greatest follower force on the cam

[35] F_min = min F — Cam follower kinematics
= scan of each segment, refined
= 20 N
Smallest follower force on the cam (negative: the follower leaves the cam)

[36] phi_max_pk = max |□| — Cam follower kinematics
= scan of each segment, refined
= 26.596 deg
Largest pressure angle

[37] Rb_min_phi = R_b that keeps □ at its limit — Cam follower kinematics
= s_0 = max(|s' − e|/tan □_max − s), R_p = √(s_0^2 + e^2)
= 32.963 mm
Smallest base-circle radius that keeps the pressure angle within its limit (roller follower)

[38] rho_min = 1/□_max — Cam follower kinematics
= pitch curve
= 44.681 mm
Smallest radius of curvature of the pitch curve

[39] u_rho = R_r / □_min — Cam follower kinematics
= 8 mm / 44.681 mm
= 0.17905
Roller radius against the smallest radius of curvature (undercut at 1 or more)

[40] u_phi = □_max / □_allowed — Cam follower kinematics
= 26.596° / 30°
= 0.88653
Largest pressure angle against the allowed one

[41] input_torque = max |F·ds/d□| — Virtual work
= virtual work, scan of each segment, refined
= 1.5345 N·m
Peak input torque by virtual work (mech.input_torque)

[42] output_motion = max |v| — Cam follower kinematics
= follower
= 0.24 m/s
Peak follower velocity (mech.output_motion)

[43] peak_accel = max |a| — Cam follower kinematics
= follower
= 4.5239 m/s²
Peak follower acceleration (mech.peak_accel)

[44] follower_force = max |F| — Cam follower kinematics
= m·a + k·s + F_pre + F_ext
= 120 N
Peak axial force of the follower on the cam (mech.follower_force)

[45] F_pin_pk = max |F|/cos □ — Cam follower kinematics
= roller contact normal force
= 121.33 N
Peak contact normal force on the cam, the load on the cam shaft's bearings (mech.F_pin_pk → M4, M5)

[46] F_spring_max = k·h + F_pre — Cam follower kinematics
= spring force at full lift, no inertia
= 120 N
Spring force at full lift (mech.F_spring_max → M7 working force)

[47] T_rms_in = √(mean T²) — Virtual work
= Simpson per segment
= 0.69759 N·m
RMS input torque over the cam revolution (mech.T_rms_in → E1 thermal sizing)

[48] P_mean = □·mean(T) — Virtual work
= Simpson per segment
= -4.375e-10 W
Mean input power over the cam revolution (mech.P_mean)

[49] f_in = n_in / 60
= rpm to Hz
= 2 Hz
Cam-shaft frequency, for the spring surge check of M7 (mech.f_in)

5. Results

SYMBOL	VALUE	DESCRIPTION
theta_sw	[0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, … (73 items)] deg	Table: cam angle
s_sw	[0, 0.0094867, 0.075117, 0.24921, 0.57669, 1.092, 1.8169, 2.7587, 3.91, 5.2492, 6.7418, 8.3428, … (73 items)] mm	Table: follower displacement
v_sw	[0, 0.0040889, 0.016077, 0.035147, 0.06, 0.088942, 0.12, 0.15106, 0.18, 0.20485, 0.22392, 0.23591, … (73 items)] m/s	Table: follower velocity
a_sw	[0, 1.1709, 2.2619, 3.1989, 3.9178, 4.3697, 4.5239, 4.3697, 3.9178, 3.1989, 2.2619, 1.1709, … (73 items)] m/s²	Table: follower acceleration

SYMBOL	VALUE	DESCRIPTION
j_sw	[170.55, 164.74, 147.7, 120.59, 85.273, 44.141, 1.0443e-14, -44.141, -85.273, -120.59, -147.7, -164.74, … (73 items)] m/s³	Table: follower jerk
phi_sw	[-12.025, -11.642, -10.507, -8.6775, -6.2746, -3.4811, -0.52955, 2.3281, 4.8597, 6.8841, 8.2863, 9.0164, … (73 items)] deg	Table: pressure angle
kappa_sw	[0.020833, 0.01767, 0.01465, 0.011975, 0.0098989, 0.0086679, 0.008423, 0.0091267, 0.010569, 0.012452, 0.014489, 0.016465, … (73 items)] 1/mm	Table: curvature of the pitch curve (the radius of curvature is 1/κ where κ is positive)
F_sw	[20, 20.282, 20.828, 21.886, 23.667, 26.334, 29.989, 34.667, 40.334, 46.886, 54.161, 61.948, … (73 items)] N	Table: follower force on the cam
T_in_sw	[0, 0.0065993, 0.026647, 0.061213, 0.113, 0.18639, 0.28638, 0.41673, 0.57774, 0.76432, 0.96511, 1.163, … (73 items)] N·m	Table: input torque by virtual work
seg_start	[0, 120, 180, 300] deg	Segments: start angle
seg_beta	[120, 60, 120, 60] deg	Segments: angle
seg_s_end	[20, 20, 0, 0] mm	Segments: displacement at the end
seg_v_pk	[0.24, 0, 0.225, 0] m/s	Segments: peak velocity
seg_a_pk	[4.5239, 0, 4.1569, 0] m/s²	Segments: peak acceleration
seg_j_pk	[170.55, 0, 259.2, 0] m/s³	Segments: peak jerk
seg_law	"1: rise (Cycloidal); 2: dwell; 3: return (3-4-5 polynomial); 4: dwell"	Segments: kind and motion law
lift	20 mm	Greatest follower displacement
v_pk	0.24 m/s	Peak follower velocity
a_pk	4.5239 m/s²	Peak follower acceleration
j_pk	259.2 m/s³	Peak follower jerk
F_max	120 N	Greatest follower force on the cam

SYMBOL	VALUE	DESCRIPTION
F_min	20 N	Smallest follower force on the cam (negative: the follower leaves the cam)
phi_max_pk	26.596 deg	Largest pressure angle
Rb_min_phi	32.963 mm	Smallest base-circle radius that keeps the pressure angle within its limit (roller follower)
rho_min	44.681 mm	Smallest radius of curvature of the pitch curve
u_rho	0.17905	Roller radius against the smallest radius of curvature
u_phi	0.88653	Largest pressure angle against the allowed one
input_torque	1.5345 N·m	Peak input torque by virtual work (mech.input_torque)
output_motion	0.24 m/s	Peak follower velocity (mech.output_motion)
peak_accel	4.5239 m/s²	Peak follower acceleration (mech.peak_accel)
follower_force	120 N	Peak axial force of the follower on the cam (mech.follower_force)
F_pin_pk	121.33 N	Peak contact normal force on the cam, the load on the cam shaft's bearings (mech.F_pin_pk → M4, M5)
F_spring_max	120 N	Spring force at full lift (mech.F_spring_max → M7 working force)
T_rms_in	0.69759 N·m	RMS input torque over the cam revolution (mech.T_rms_in → E1 thermal sizing)
P_mean	-4.375e-10 W	Mean input power over the cam revolution (mech.P_mean)
n_in	120 rpm	Cam-shaft speed, passed through for E1's load speed (mech.n_in)
f_in	2 Hz	Cam-shaft frequency, for the spring surge check of M7 (mech.f_in)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Pressure angle φ_max within the allowed maximum	u_phi = 0.88653	u_phi ≤ 1	0.887	Cam follower kinematics
PASS	No undercut: ρ_min of the pitch curve exceeds the roller radius	□_min = 44.681 mm	> 8 mm	0	Cam follower kinematics
PASS	Follower stays on the cam: F_min ≥ 0	F_min = 20 N	≥ 0 N	0	Cam follower kinematics

7. Applicability

No warnings.

8. Verification

Template mech.mechanism@1.2.0 Content hash 082569fdde676480 Engine calc-engine/1.0.0
Known-answer cases passed 16/16 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0